

1. Find the sum of each expression using the fewest terms possible.

a. $(x + 9) + (2x + 3)$

b. $(4y - 7) + (16 + 6y)$

2. Find the difference of each expression using the fewest terms possible.

a. $(11k + 15) - (7k + 3)$

b. $(8 - 2x) - (-5x + 4)$

3. Complete each statement.

- a. The

- ☐ associative
 - ☐ commutative
 - ☐ distributive

 property can be used

to state that $(d + 4)8 + 7d$ is equivalent to $8d + 32 + 7d$.

- b. The

- ☐ associative
 - ☐ commutative
 - ☐ distributive

 property can be used

to state that $\frac{6}{5}x + \left(\frac{3}{4}x + \frac{4}{5}\right)$ is equivalent to $\left(\frac{3}{4}x + \frac{4}{5}\right) + \frac{6}{5}x$.

4. Write each expression using the fewest possible terms.

- a. $14.8 - (6.3 - 2.9r)$ b. $(5.9x - 1.7 - 3.76x) + (14.78 + 16.3x)$

5. What is the result when $\frac{7}{8}w + \frac{5}{6}$ is subtracted from $\frac{14}{4}w - \frac{23}{6}$?

6. What is the result when $8\frac{9}{20}x - 7\frac{3}{5}$ is added to $11\frac{1}{4} - 9\frac{3}{5}x$?

7. Use the given expressions to answer each.

$$14(2m + 5) - 2 \text{ and } 28m + 3$$

- a. Choose a value for m to determine if the expressions are equivalent. Show your work.

- b. Use properties of operations to determine if the expressions are equivalent. Show your work.

- c. Complete the statement.

The expressions ☐ are
☐ are not equivalent.

8. Write an equivalent expression only using the property of operations given.

Expression	Property	Equivalent Expression
$(5 - 0.7c) + (1.2c - 4.8)$	Commutative	
$12y + (10y + 4) - 17$	Associative	
$11x - 7(6 - 9x) + 23$	Distributive	